KYAMBOGO UNIVERSITY

P. O. BOX 1, KYAMBOGO – KAMPALA, UGANDA

School of computing and information science

Department of Computer Science

COURSE UNIT: INTRODUCTION TO DATA COMMUNICATION & NETWORKING

PROGRAM: BACHELOR OF INFORMATION TECHNOLOGY AND COMPUTING

COURSE CODE: SCS 2103

YEAR: 2

SEMESTER: 1

STUDENT NAME: CHELANGAT MARY

REGISTRATION NO: 24/U/ITE/04619/PE

LECTURER'S NAME: MR.SAMUEL EIBU

ASSIGNMENT AND COURSEWORK 1

Project 1:

ABA Company is setting up an office in Kampala, and they want to establish a Local Area Network (LAN) where they can share one central printer and access files on a network shared folder on the server. The organization has acquired 4 new desktop computers and 3 laptop computers, and all should be networked so they can access shared resources, such as one central printer that is to be placed in a common area, and a shared network folder.

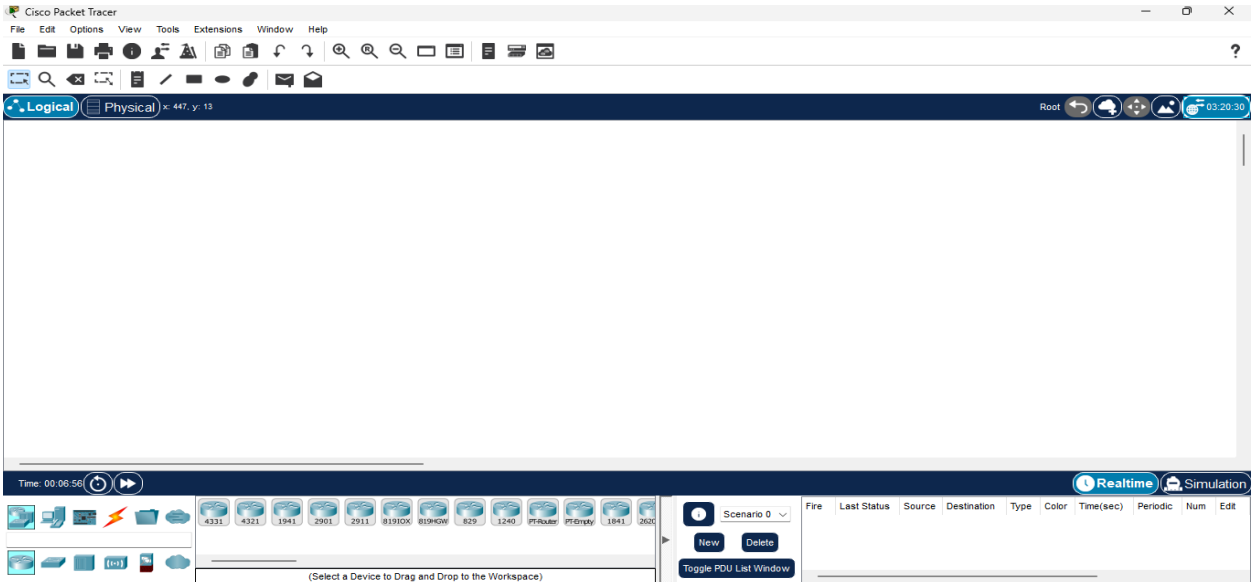
Four desktop computers are to be directly connected by an Ethernet cable, and 3 laptop computers are to be connected by one (1) wireless access point.

The ABA has a consultant who is guiding the network administrator on the network design and has advised that the organization should use the local private network of IP address: 192.168.1.0/24 for its LAN setup.

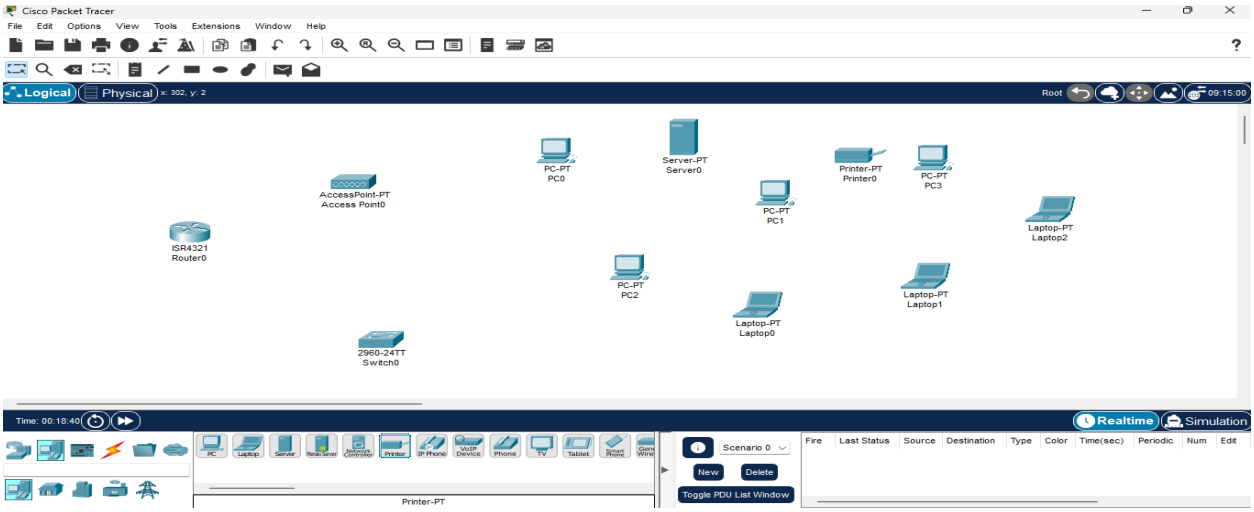
ABA has decided that they will be using Lets Connect as their Internet service provider to offer them internet. Let's Connect has offered them the public IP address 102.223.89.106 with the subnet mask 255.0.0.0 to be configured on their router.

A) **SET UP THE LAN NETWORK FOR ABA WITH PHYSICAL AND LOGICAL TOPOLOGIES IMPLEMENTED AS GUIDED ABOVE IN THE PROJECT. (TAKE SCREENSHOTS OF YOUR OUTPUT FOR EVERY STEP PERFORMED)**

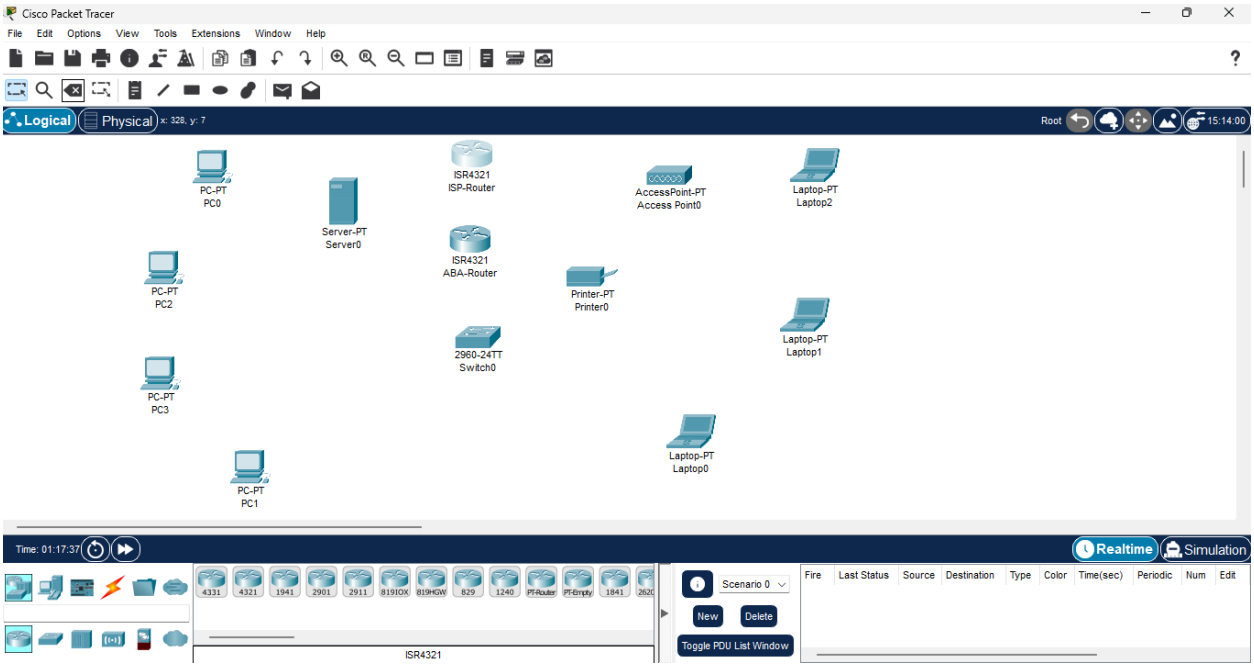
(i) Here I opened up the Cisco Packer Tracer Software.



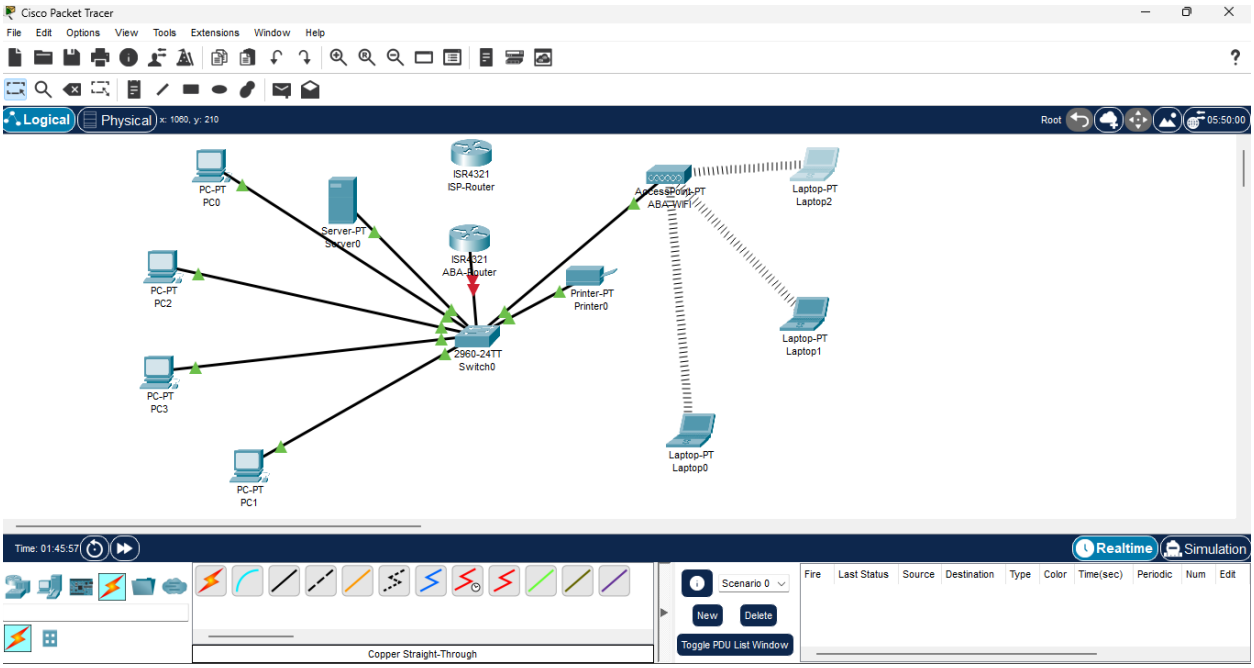
(ii) Now I have dragged all the necessary devices on to the Packer tracer dashboard



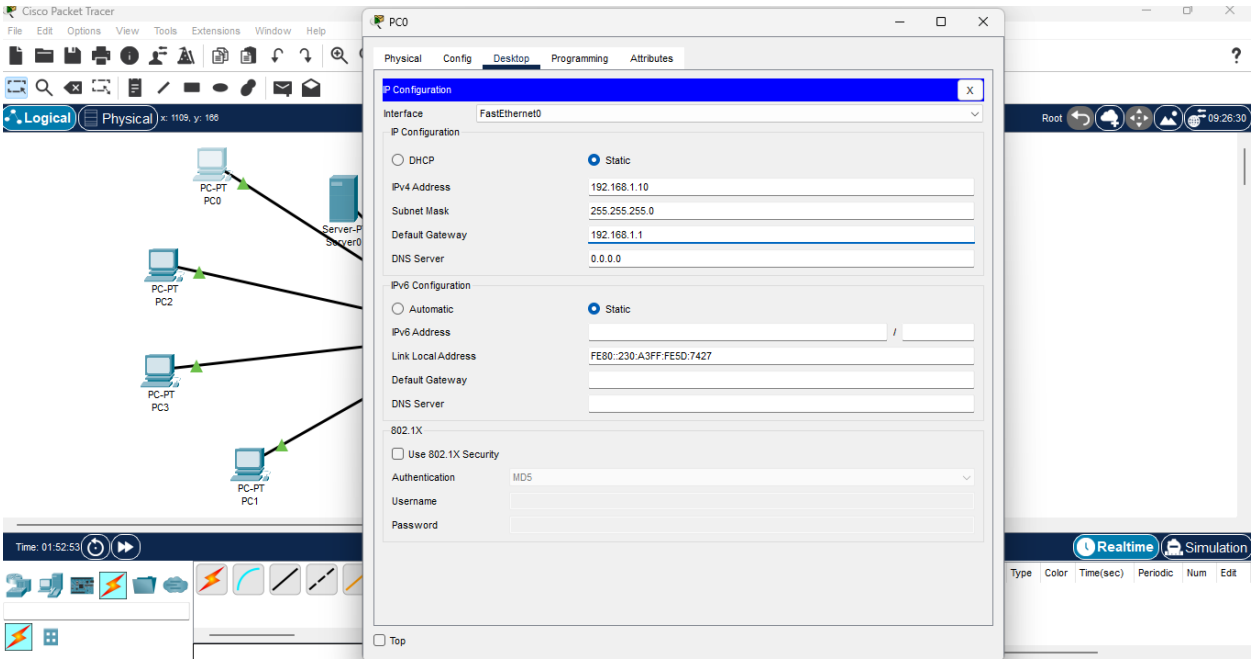
(iii) Now I have arranged the devices



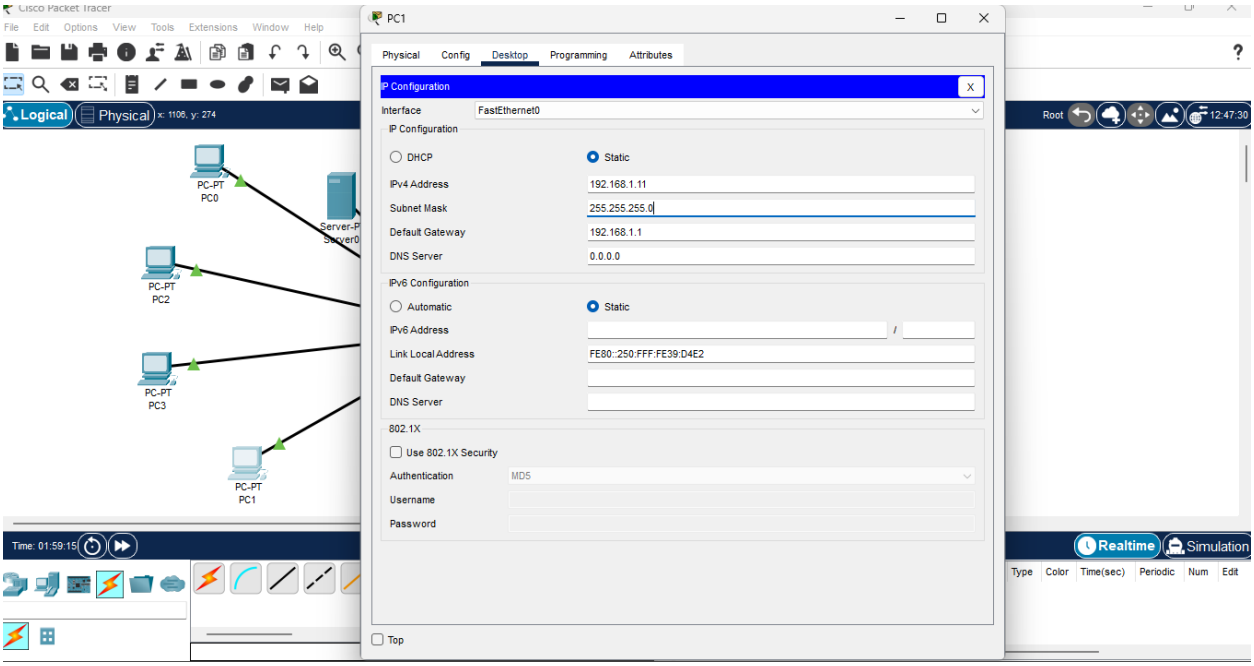
(iv) Now I have connected the different devices to their respective devices and changed the names to match the question



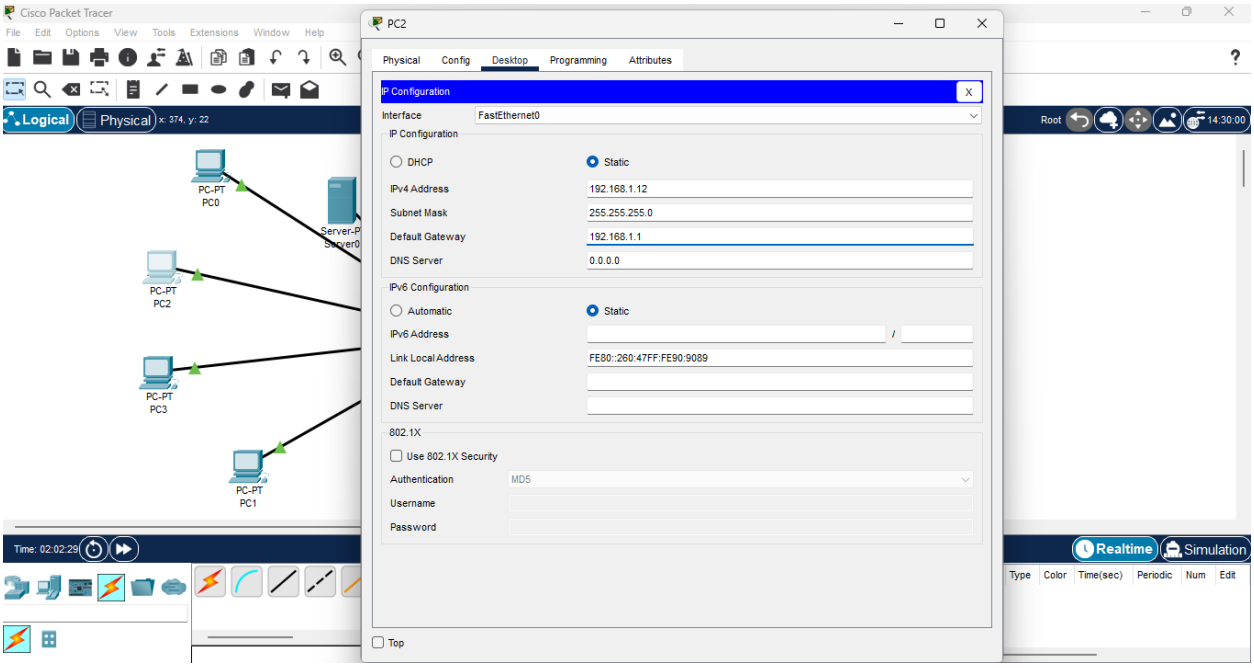
(v) Configuring the IP address for desktop 0



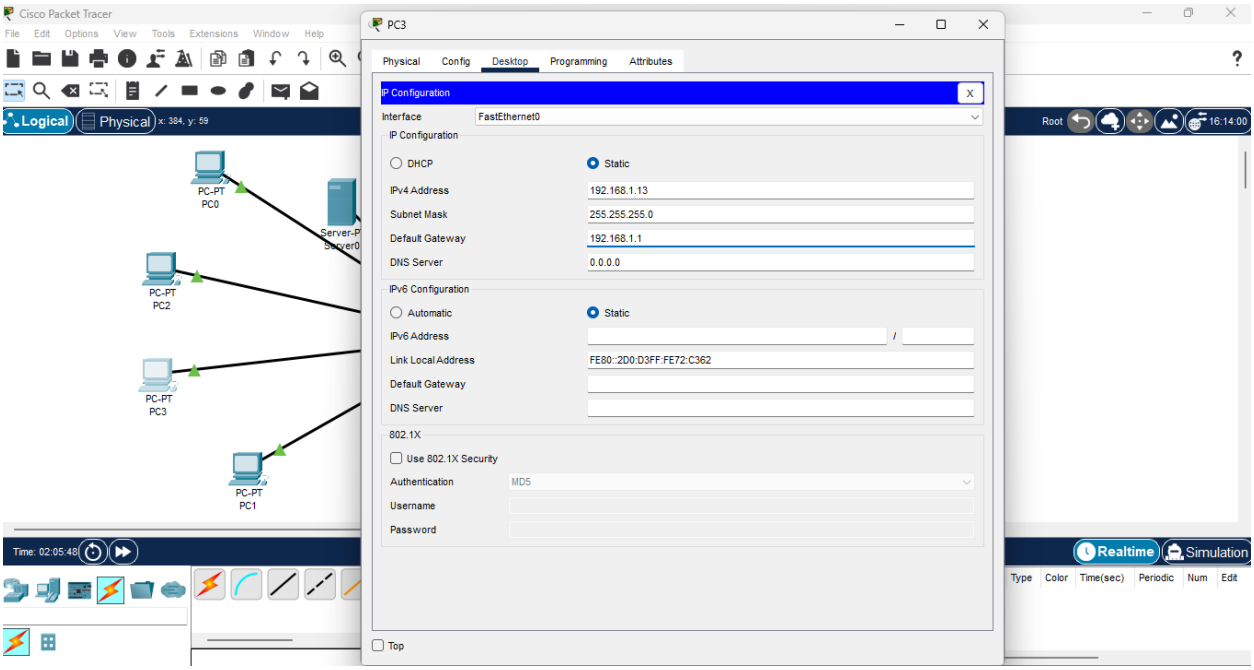
(vi) Desktop 1



(vii) Desktop 2

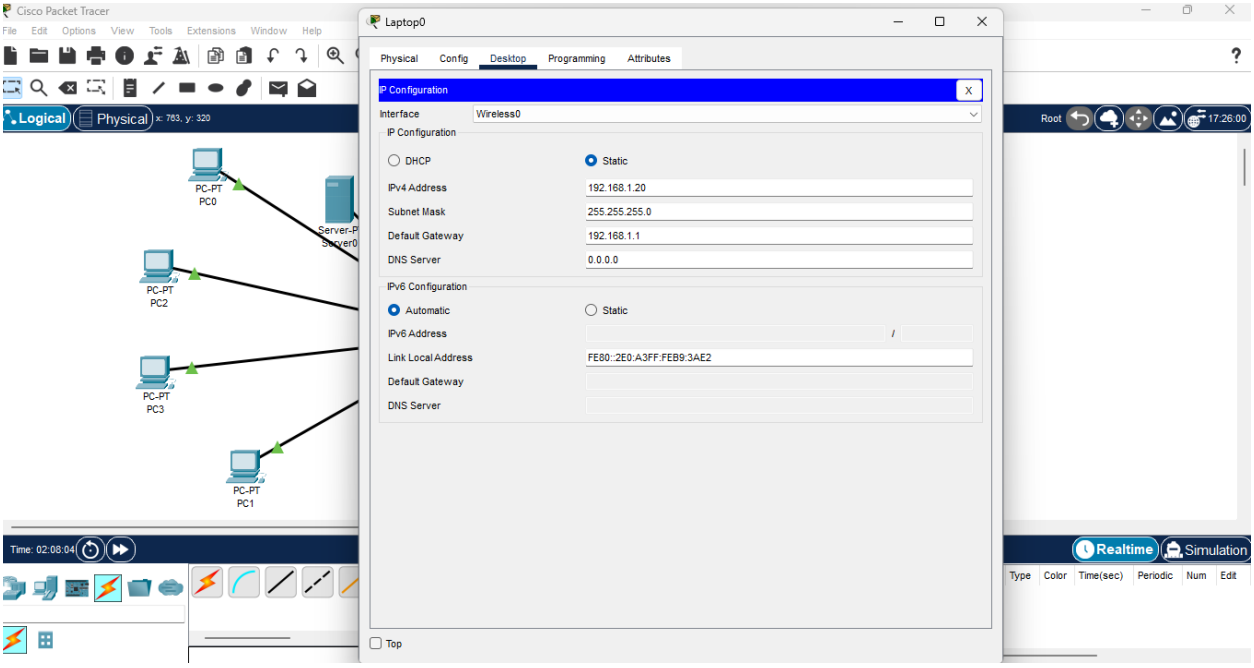


(viii) Desktop 3

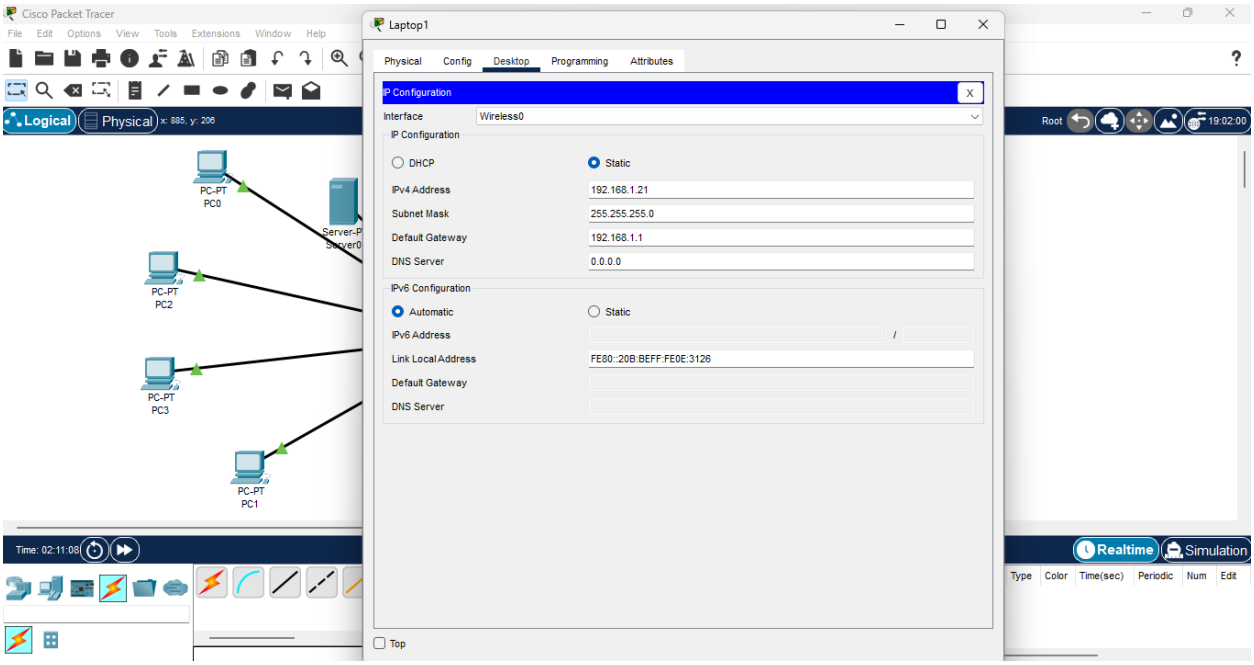


(xi) Configuring the IP address for the Laptops

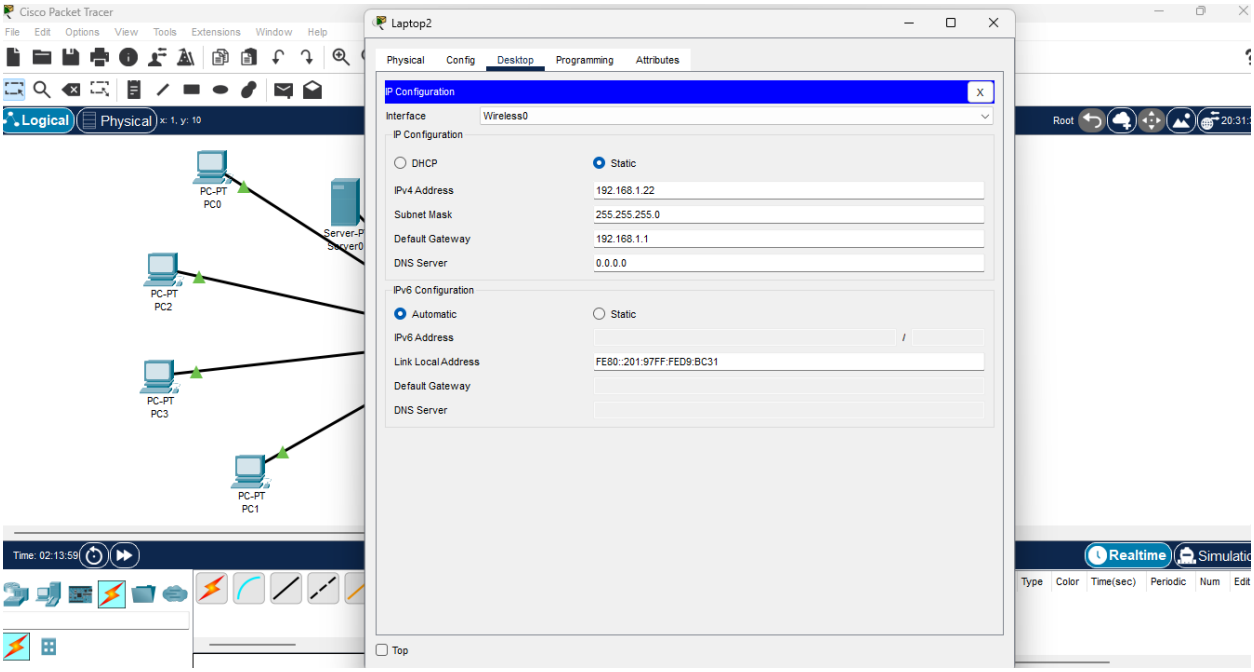
Laptop 0



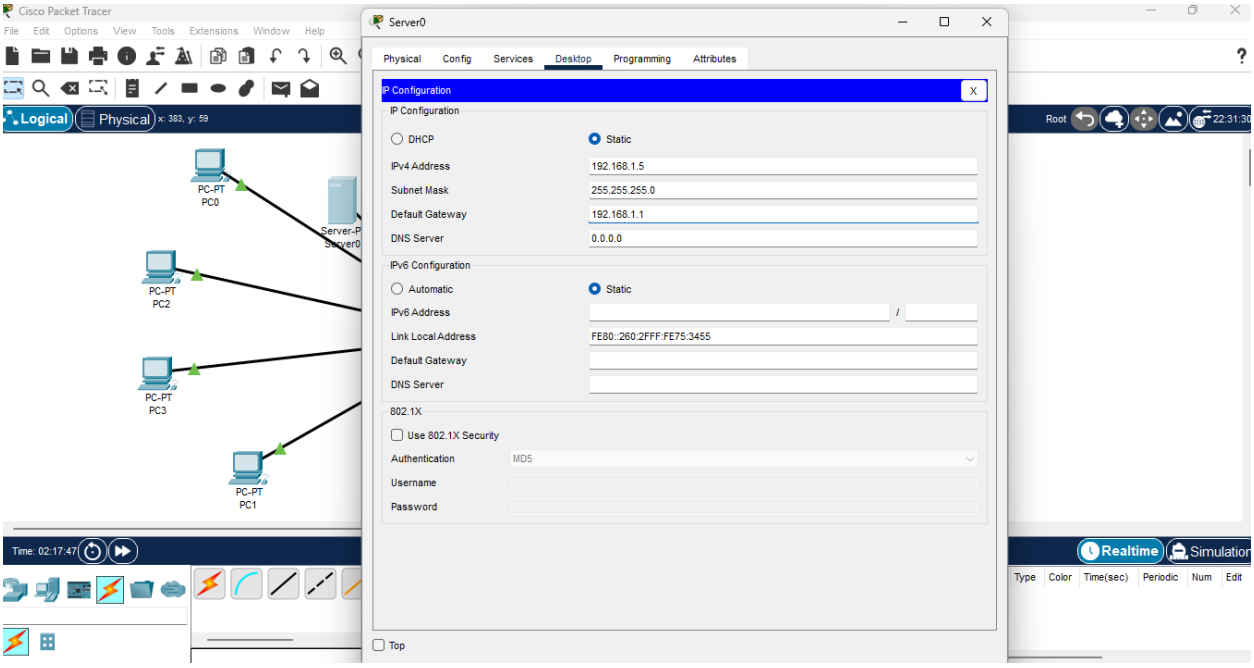
Laptop 1



Laptop 2



Server IP address configuration



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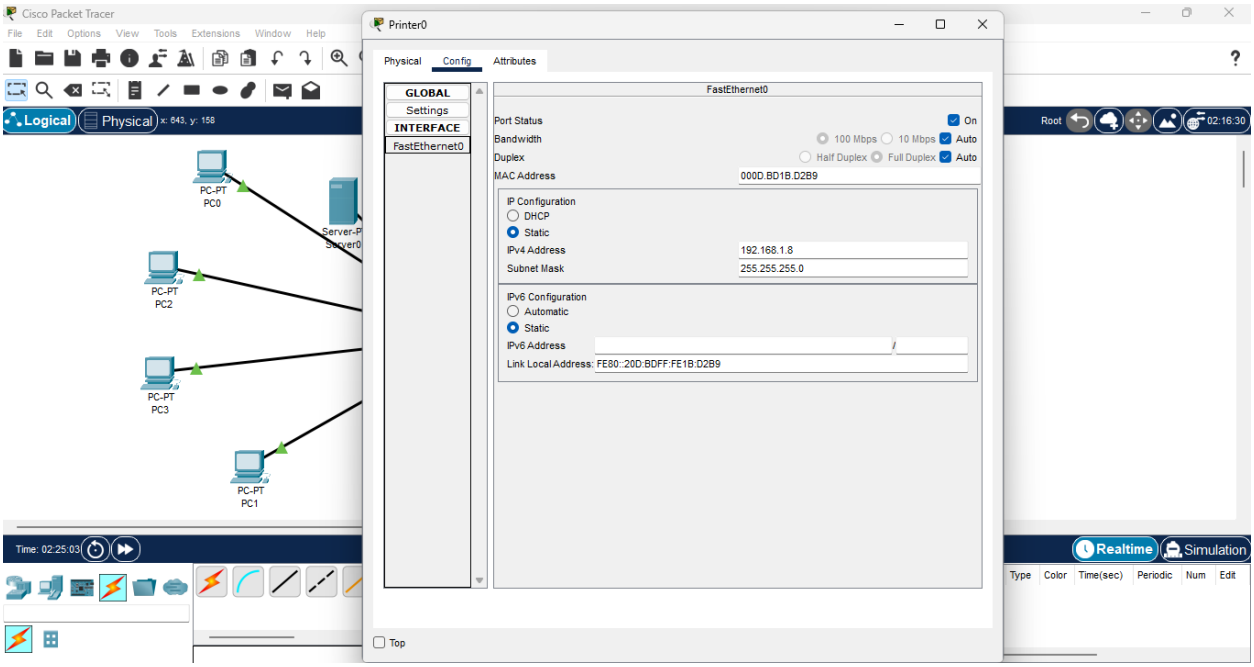
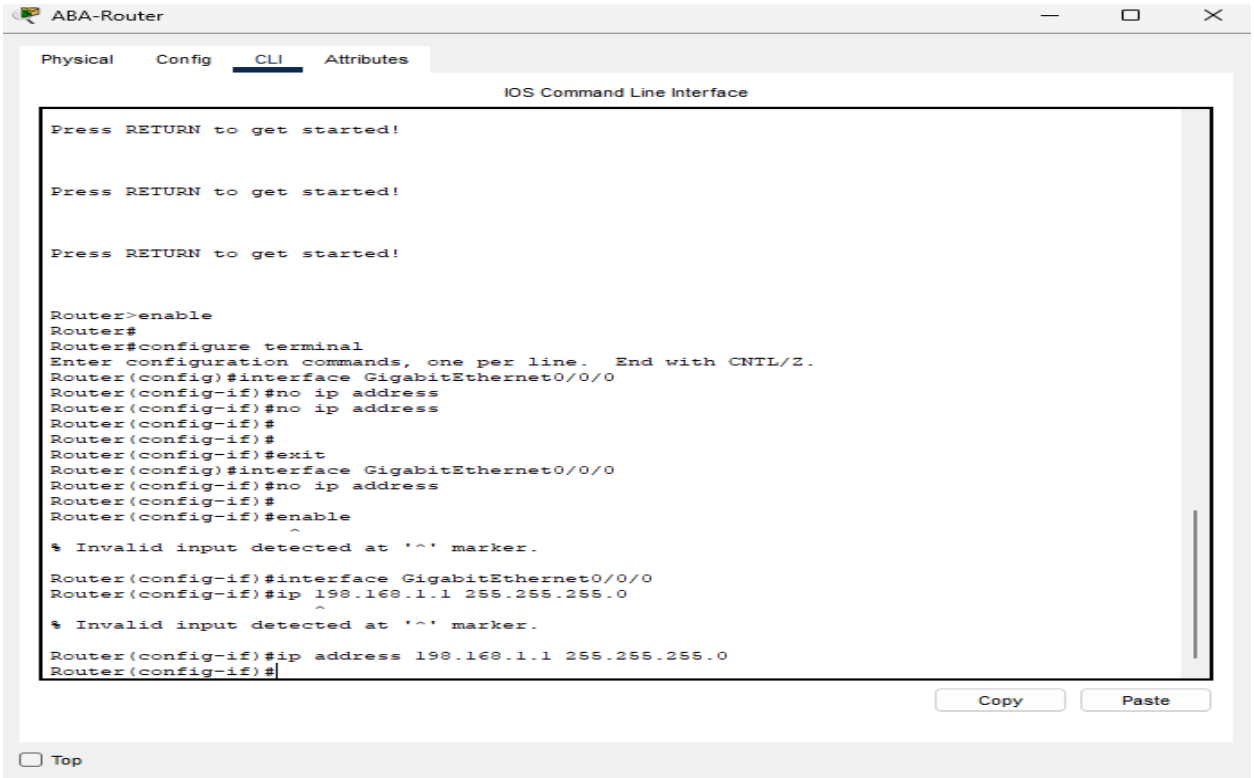
Printer IP address configuration



ABA router IP address configuration



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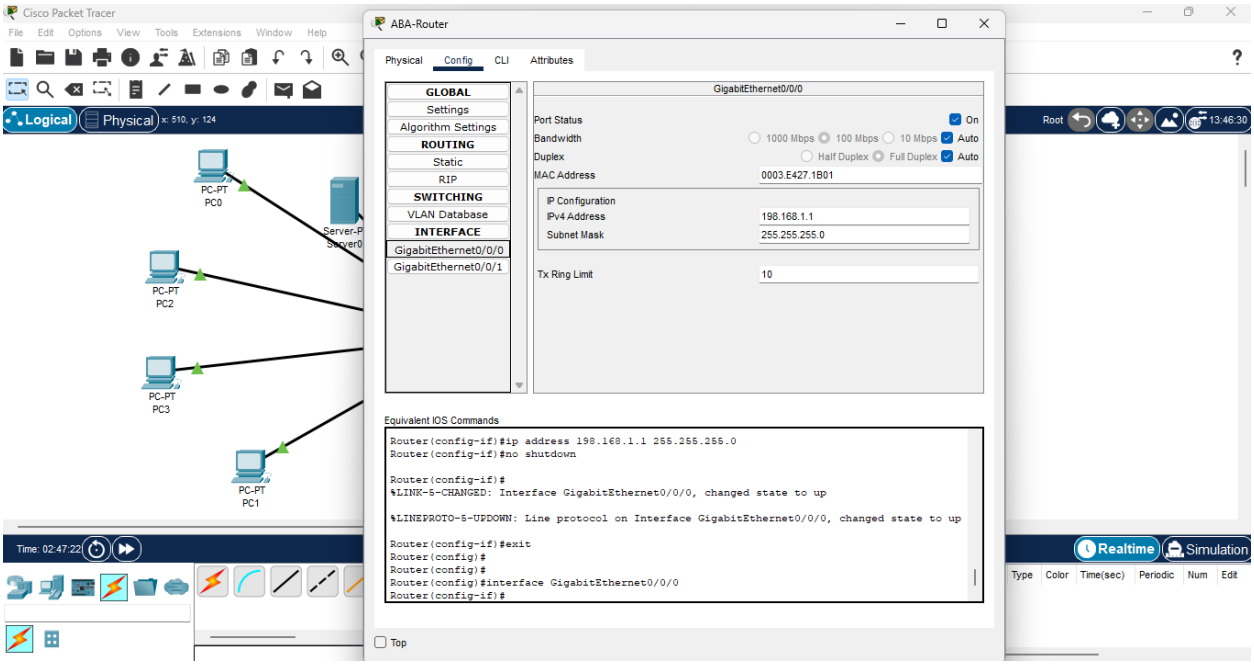
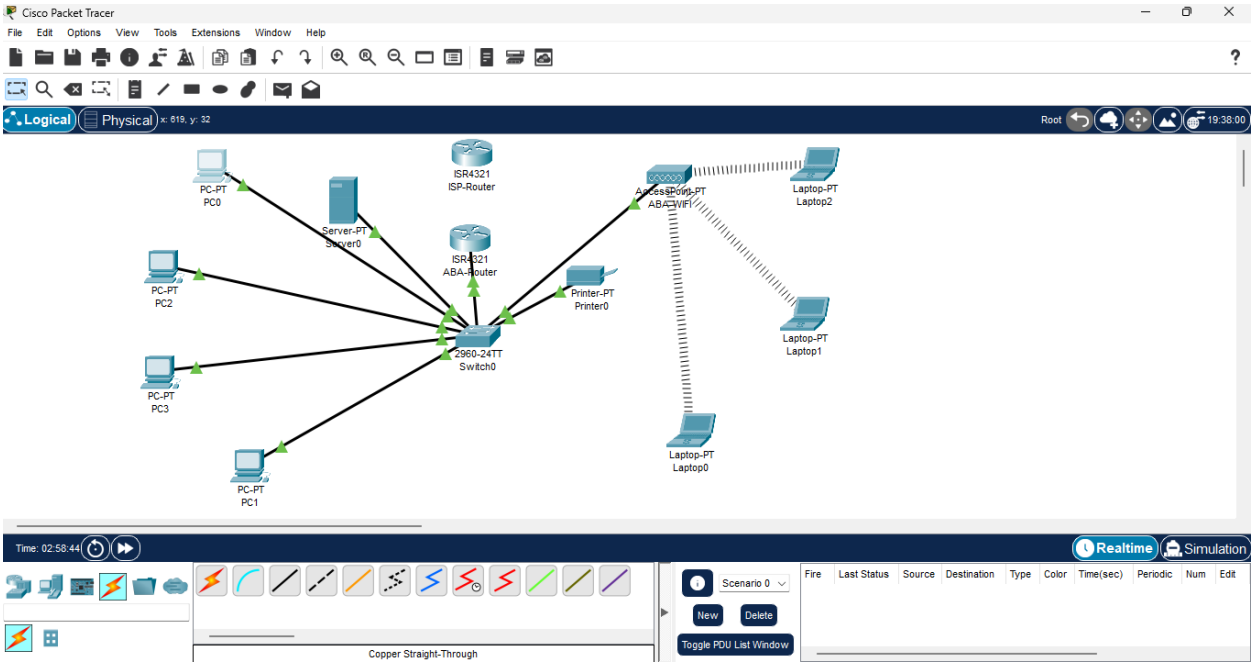
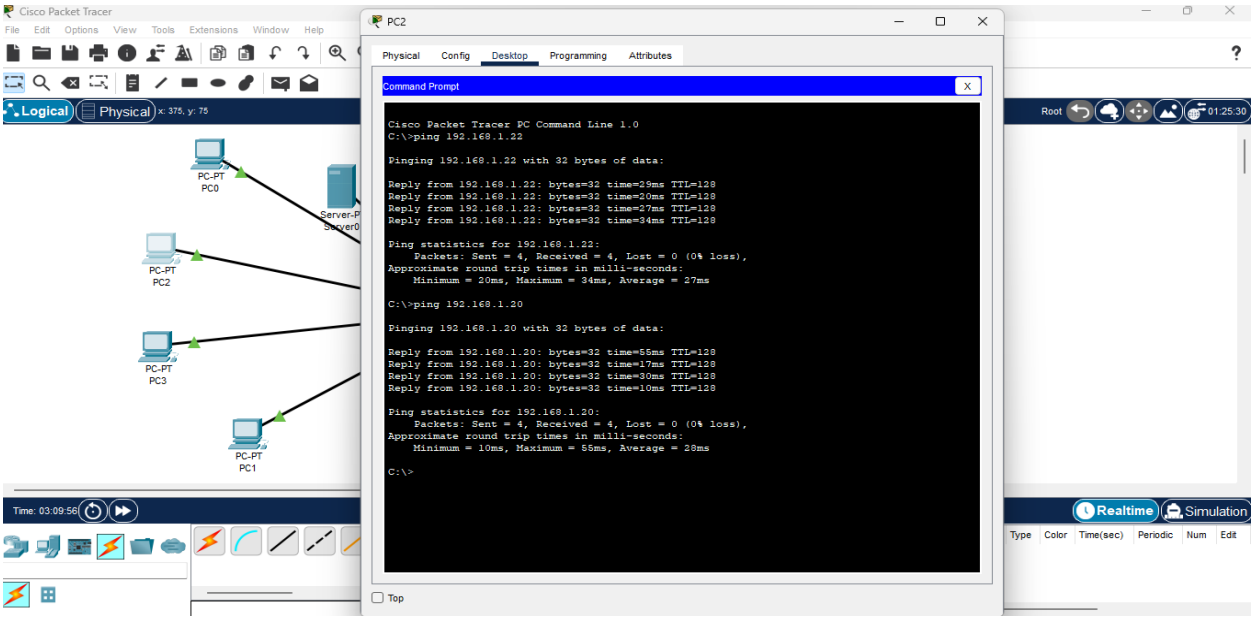


(ix) The Final LAN Network for ABA

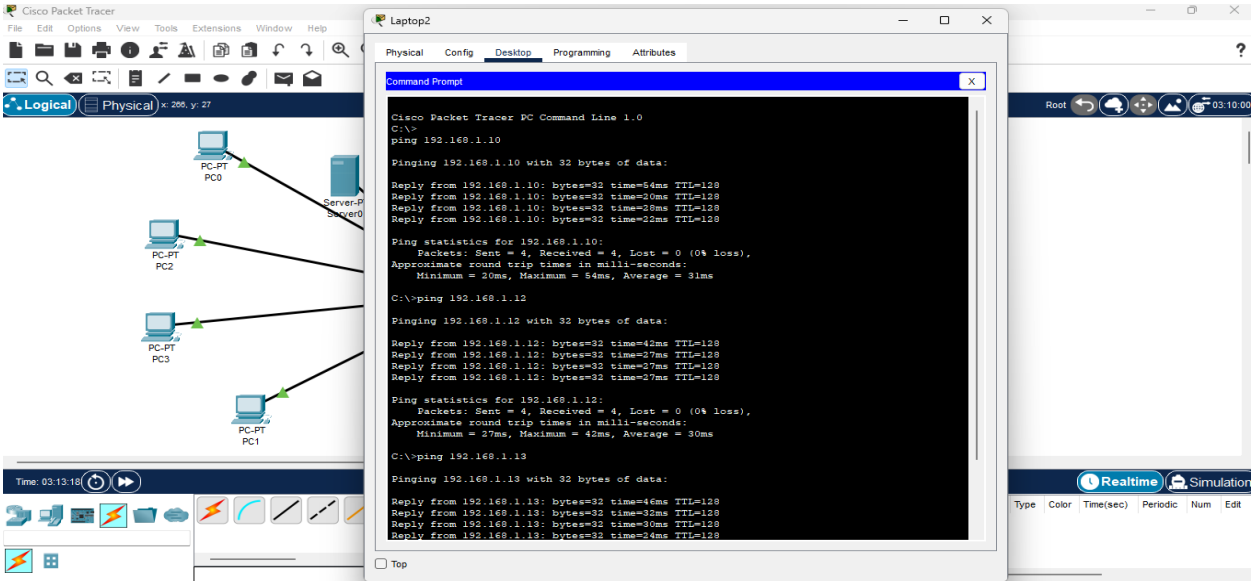


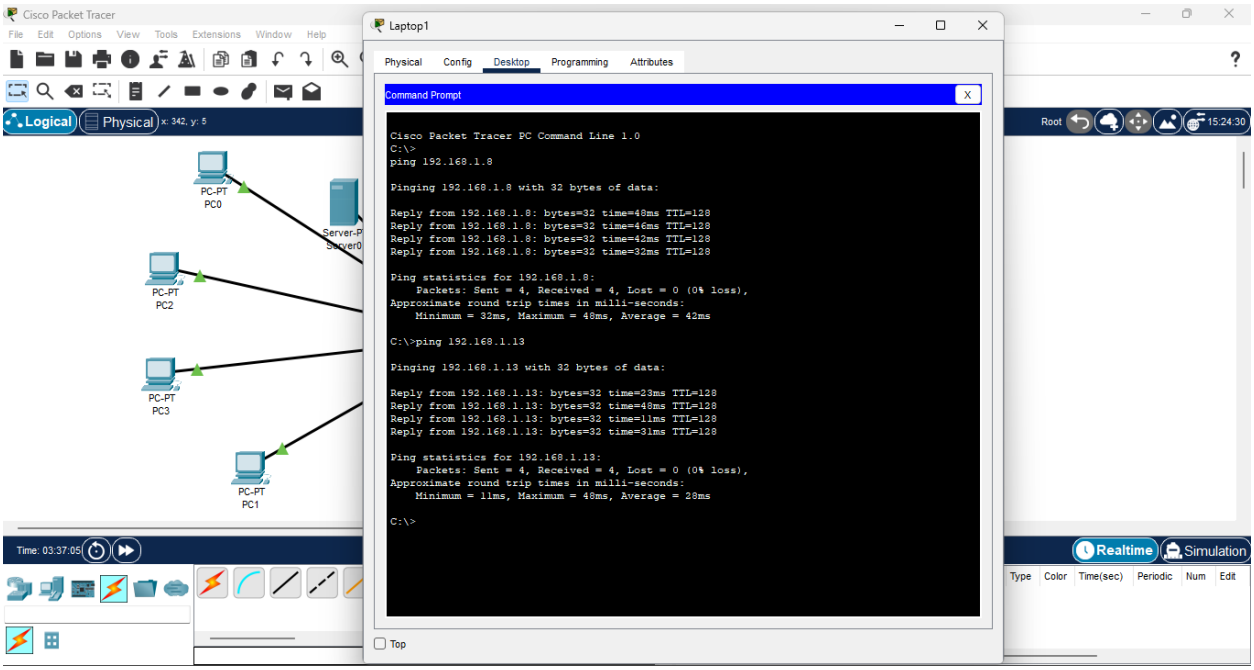
B) TEST IF DESKTOP COMPUTERS CAN COMMUNICATE WITH LAPTOP COMPUTERS

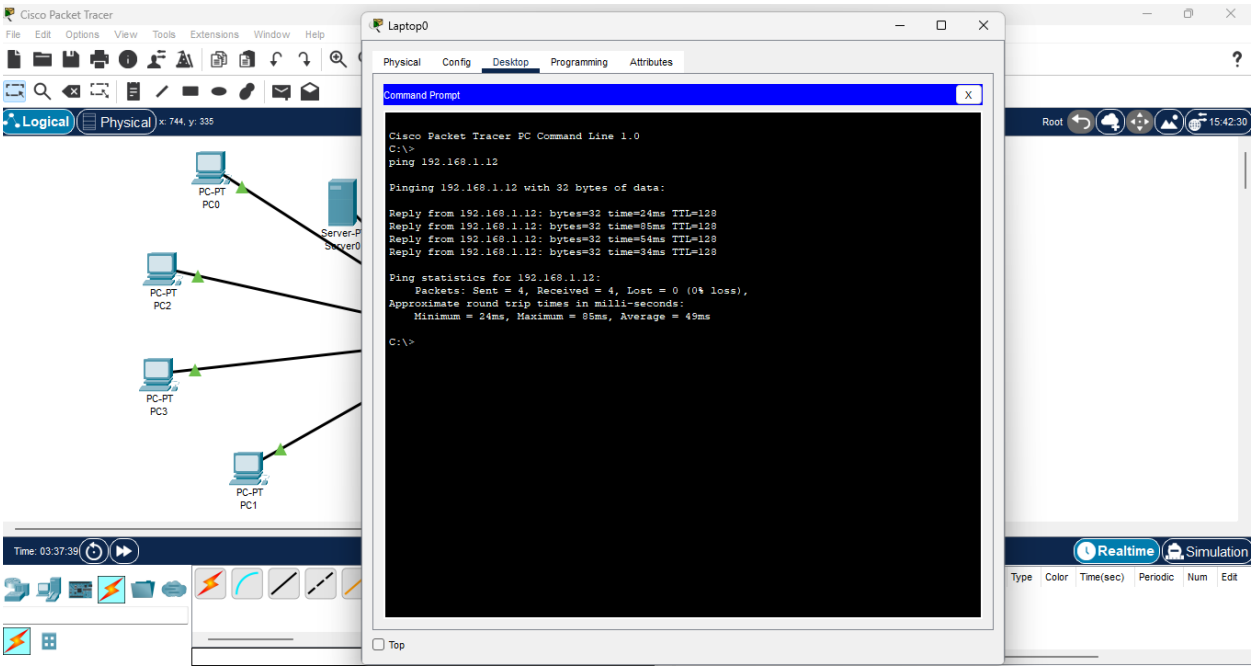
(i) The Desktop computers can communicate to Laptops



(ii) The Laptops can communicate to Desktop computers







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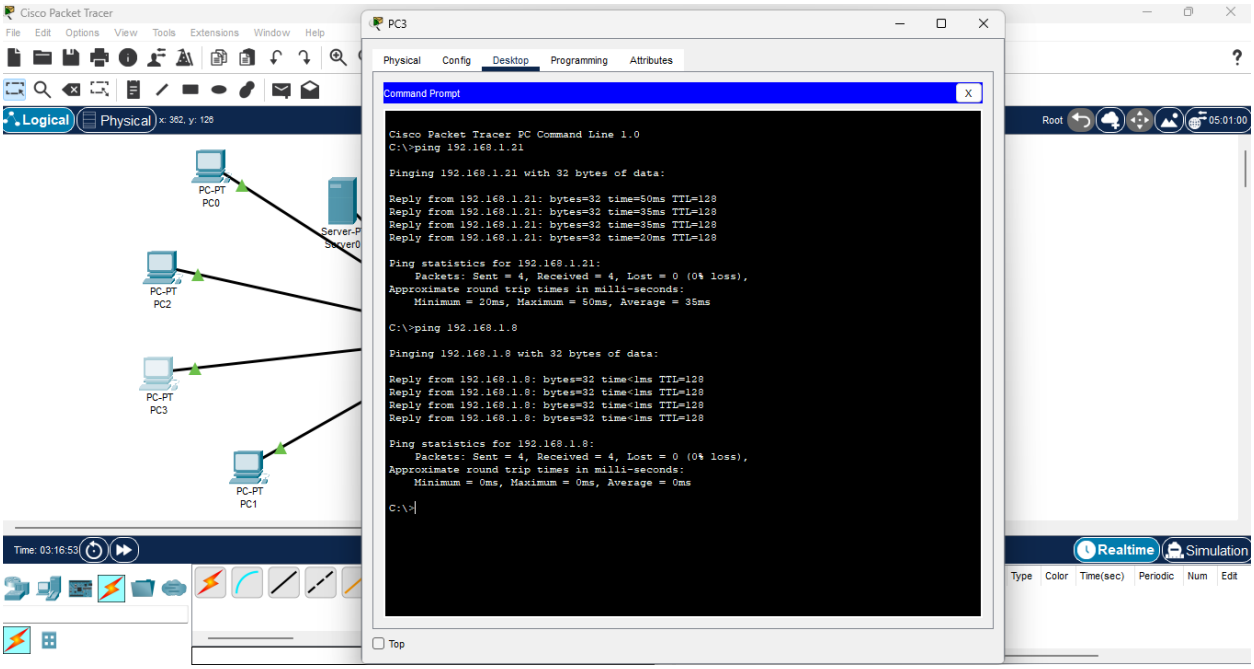
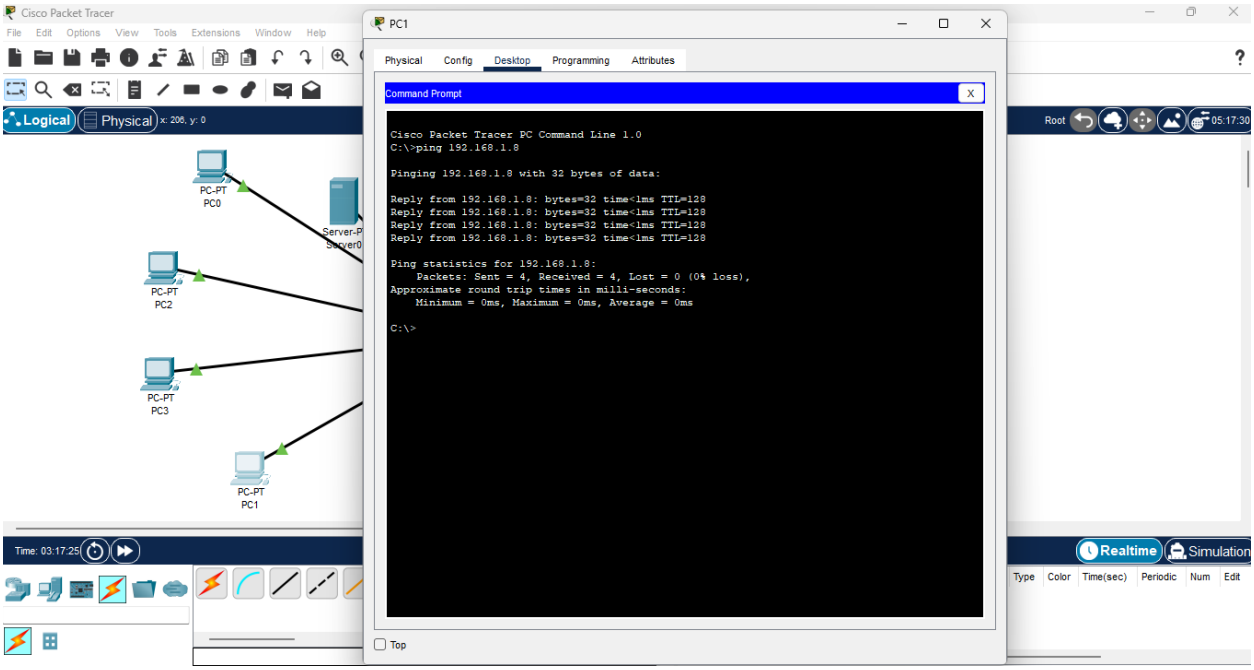
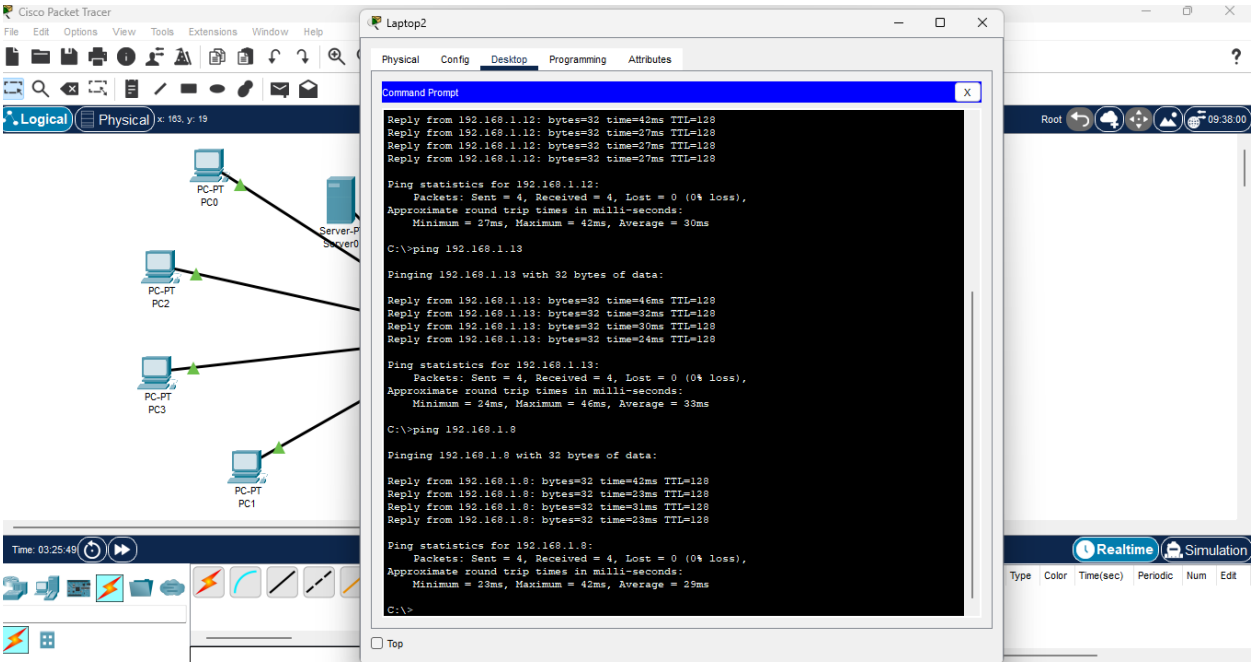
C) TEST IF ALL COMPUTERS (BOTH DESKTOP AND LAPTOPS) CAN ACCESS THE CENTRAL PRINTER

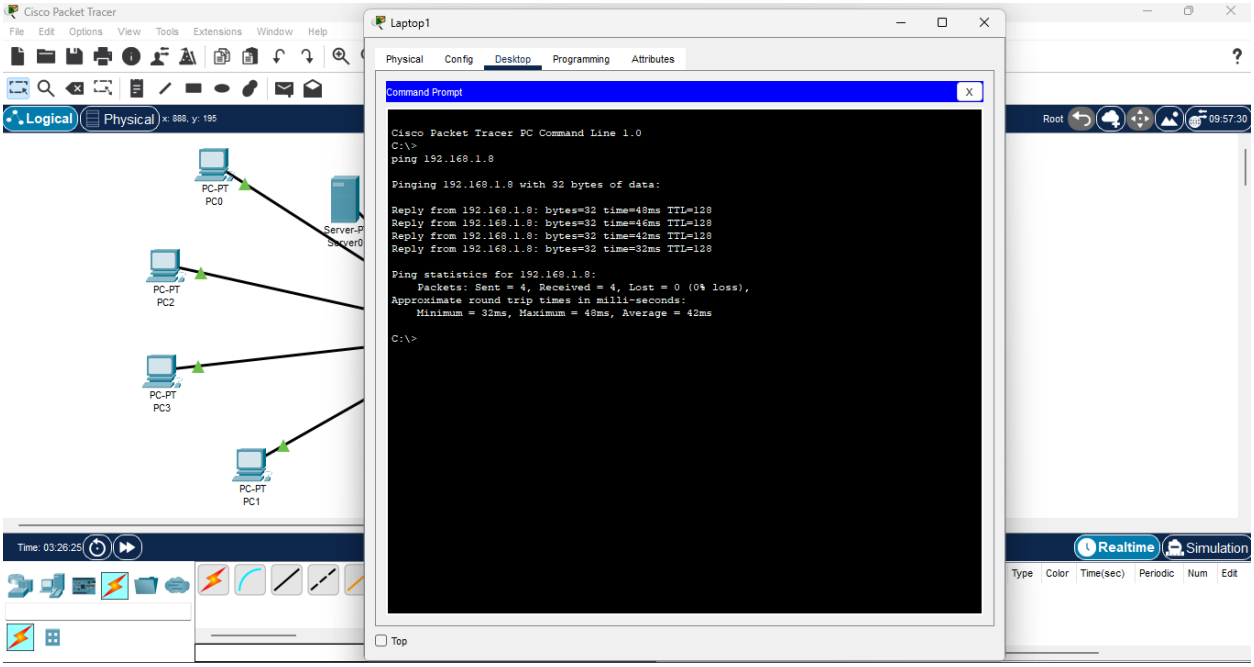
(i) The Desktop Computers can access the central printer





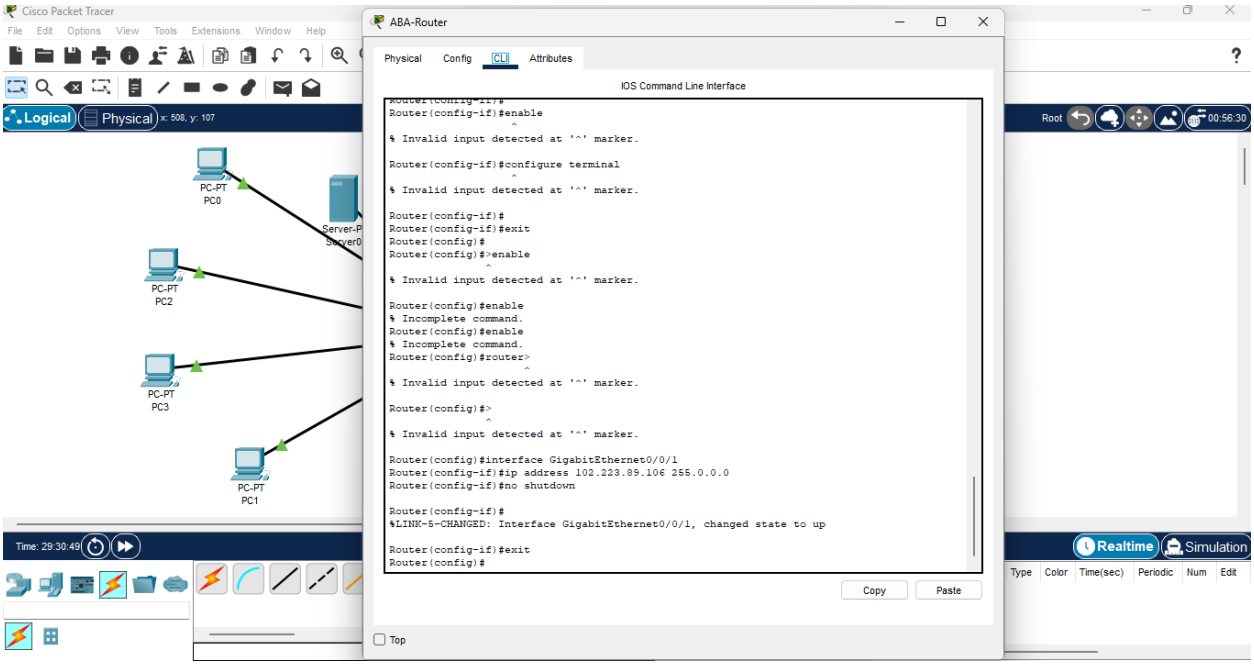
(ii) The Laptops can access the Central Printer

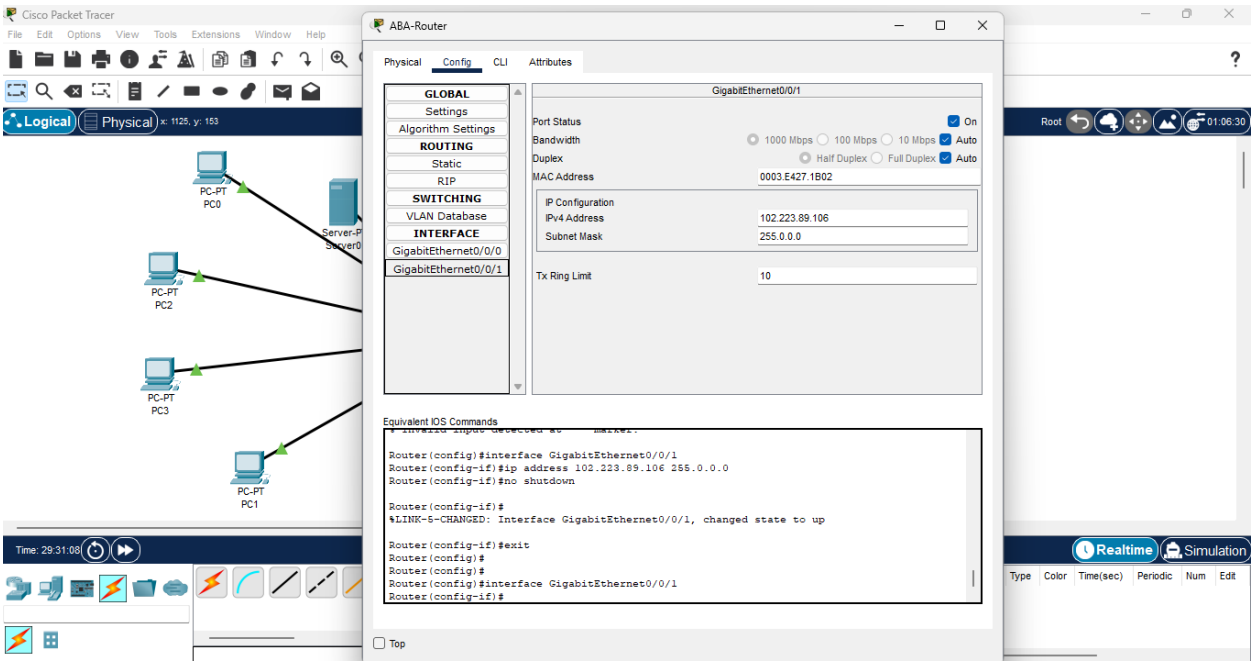




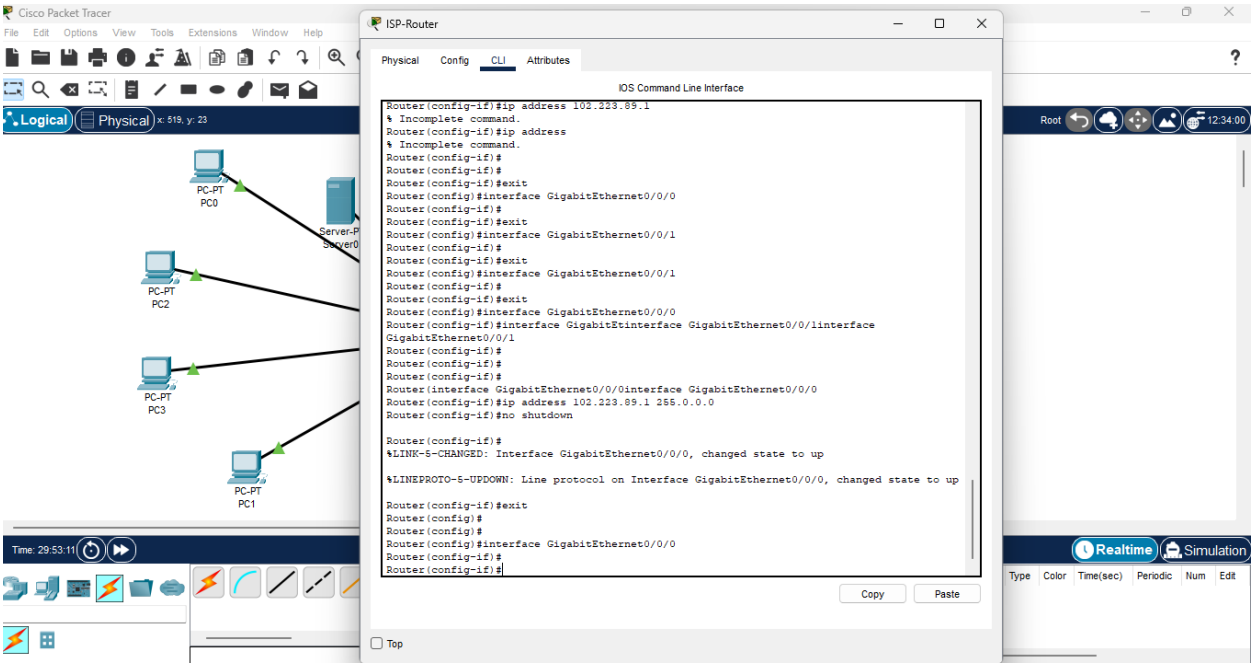
D) CONFIGURE BOTH THE PUBLIC IP ADDRESS GIVEN BY THE INTERNET SERVICE PROVIDER AND THE LOCAL IP ADDRESS ADVISED BY THE CONSULTANT ON THE ROUTER IN THE NETWORK SETUP IN A) ABOVE.

(i) Configuring the ABA router IP address on the public side

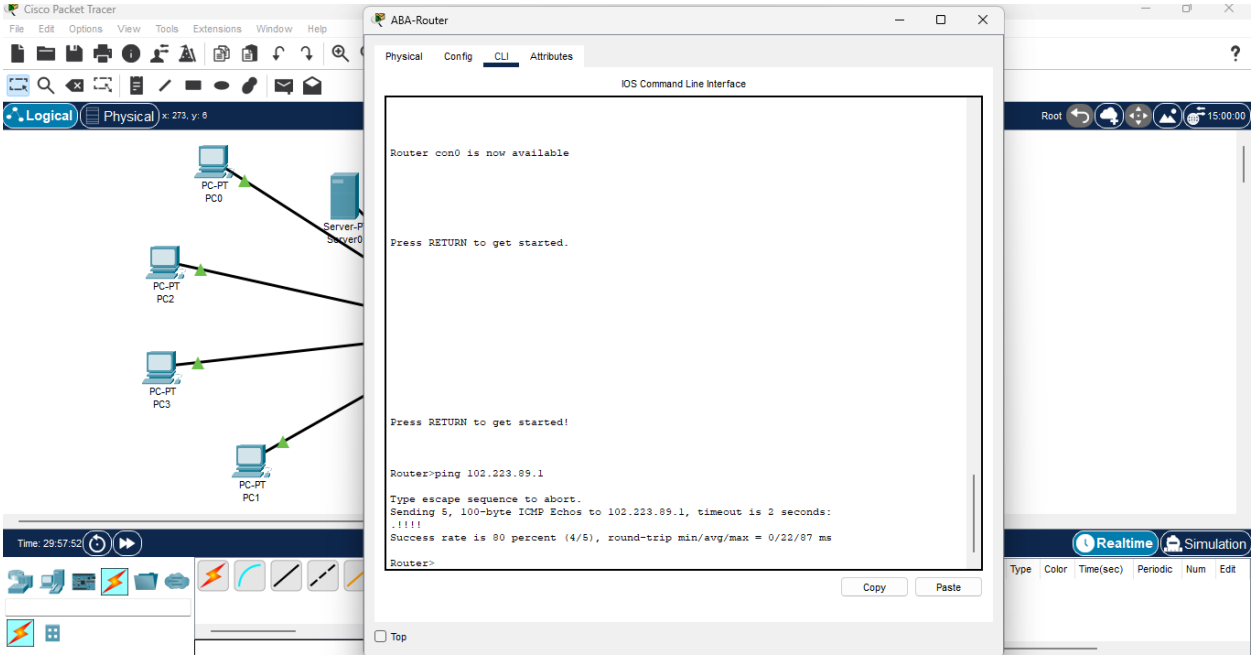




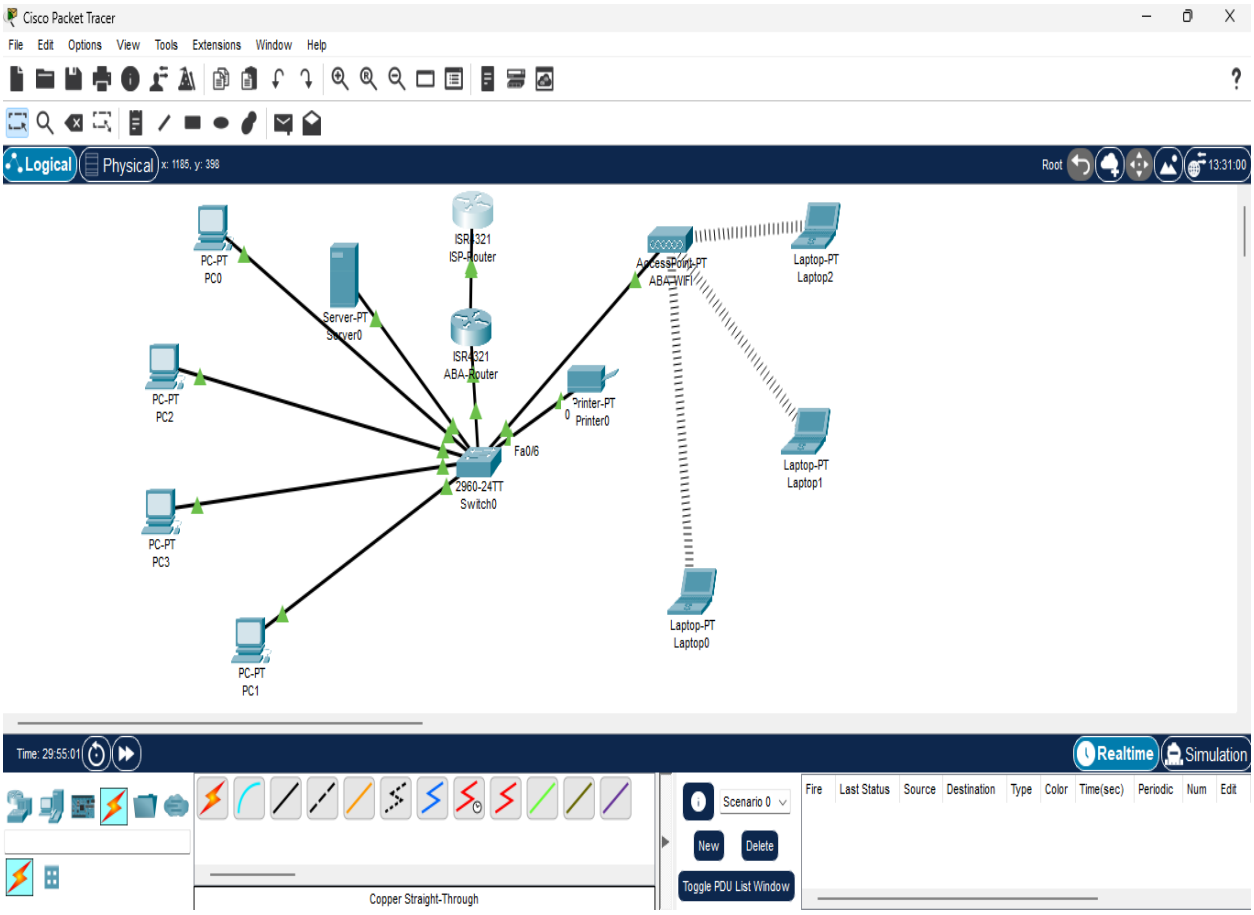
(iii) Configuring the IP address of the ISP router



(iv) The Public Routing is successful.



(v) The Final LAN- Network Topography



ASSIGNMENT TASK 2:

CONDUCT RESEARCH AND WRITE A REPORT/ESSAY WITH THE AID OF RECOMMENDED INTRODUCTION TO DATA COMMUNICATION AND NETWORKING BOOKS, JOURNALS, ARTICLES, AND ONLINE UP-TO-DATE MATERIALS AND PRACTICAL EXAMPLES FOR THE FOLLOWING:

a) THE ROLE OF THE DATA LINK LAYER IN NETWORKING AND DATA COMMUNICATION

- The data link layer is essential for enabling reliable communication within a local network. According to Kurose and Ross (2017), the data link layer provides framing, which is the process of taking raw bits from the physical layer and organizing them into meaningful data units called frames. These frames help devices identify where each message begins and ends, ensuring accurate interpretation of transmitted information.
- Another key role of the data link layer is error detection and correction. Because physical media such as copper cables or wireless channels can introduce noise, the data link layer uses mechanisms such as cyclic redundancy checks (CRC) to detect corrupted frames and request retransmission. This significantly increases the reliability of communication (Stallings, 2020).
- The data link layer also performs media access control (MAC), which determines how devices share the transmission medium. For example, Ethernet networks use Carrier Sense Multiple Access with Collision Detection (CSMA/CD) to manage simultaneous transmissions and avoid data collisions. Without MAC functions, multiple devices would transmit at the same time, causing interference and communication breakdown.
- Additionally, the data link layer handles physical addressing through MAC addresses. Each network device has a unique MAC address burned into its hardware. Switches use these MAC addresses to forward frames accurately within the LAN. Unlike IP addresses (Layer 3), MAC addresses do not change, making them useful for local device identification.

In summary, the data link layer ensures reliable communication through framing, error control, MAC functions, and physical addressing. These roles make it foundational for effective data communication within local networks.

B) EXPLAIN HOW LAN INTERCONNECTION CAN BE ESTABLISHED WITH PRACTICAL EXAMPLES, AND PRACTICAL SIMULATION/DEMONSTRATION USING CISCO PACKET TRACER SOFTWARE.

LAN interconnection refers to the process of linking multiple devices such as computers, printers, switches, and routers within a limited geographical area to enable communication and resource sharing. A LAN allows users to share files, printers, internet connectivity, and application services.

A practical example of LAN interconnection involves an office environment with desktop computers, laptops, and a central network printer. Devices are connected using Ethernet cables through a switch, while wireless devices access the network through an access point. Each device is assigned a unique IP address, such as 192.168.1.x, along with a subnet mask and a default gateway, enabling communication inside and outside the LAN.

Cisco Packet Tracer provides a practical simulation platform for designing and testing LAN interconnections. To build a LAN in Packet Tracer, networking devices such as switches, routers, PCs, laptops, and servers are dragged into the workspace. A physical topology is created by connecting devices using straight-through cables for PC-to-switch connections and appropriate cabling for other links.

After building the topology, logical configuration begins. PCs are configured manually by navigating to Desktop IP Configuration, where an IP address, subnet mask, and default gateway are entered. For example, a PC might be assigned:

- IP address: **192.168.1.10**
- Subnet mask: **255.255.255.0**
- Default gateway: **192.168.1.1**

Router configuration is done through the Command Line Interface (CLI). A typical setup includes enabling interfaces, assigning IP addresses, and activating the interfaces using the no shutdown command:

```
enable
configure terminal
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
```

Once configuration is complete, connectivity is tested using the ping command. A successful ping indicates that the devices are properly interconnected. Packet Tracer also provides a Simulation Mode to visualize packet flow across the network, making troubleshooting easier.

In conclusion, LAN interconnection enables organizations to share resources efficiently. Using Cisco Packet Tracer, students and network professionals can practice building LANs,

configuring devices, assigning IP addresses, and testing connectivity in a real-world scenario without needing physical equipment.

REFERENCES (APA STYLE)

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